A1: Symmetries and conservation laws

Worked solutions

Training dynamics, a self-study course · written by Claude for Fan Pu

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`reference numbers: solutions/results.json · figures: solutions/figures/ · equation numbers (n) refer to the handout`

Every derivation here is complete; read only as far as you need. "Measured" numbers come from running each check script with the reference implementations (`--solutions`) on CPU; your numbers should agree within the tolerances stated in each problem. Diff yours with `uv run python -m tdyn.results a1-symmetries/out/results.json a1-symmetries/solutions/results.json`. Equation numbers such as (2) refer to the handout; this document numbers its own equations (S1), (S2), … Every formula below was also checked symbolically (sympy) or numerically before it was written down.

1 Problem (`noether`): Rescaling symmetry and the conserved imbalance

(a) The generator of (2) is $X(a,b) = \partial_\alpha(\alpha a, b/\alpha)|_{\alpha=1} = (a, -b)$. Differentiating $L(\alpha a, b/\alpha) = L(a,b)$ at $\alpha = 1$ by the chain rule:

$$\langle \nabla L(\theta), X(\theta)\rangle = a\,\partial_a L - b\,\partial_b L = 0 \quad \text{for every } \theta. \tag{S1}$$

Along gradient flow, with $Q = a^2 - b^2 = c$,

$$\dot{Q} = 2a\dot{a} - 2b\dot{b} = -2(a\,\partial_a L - b\,\partial_b L) = -2\langle \nabla L, X\rangle = 0. \tag{S2}$$

Note $\nabla Q = 2X$: the conserved quantity is the function whose gradient is the generator, which works because X here is itself a gradient field (that is the "Noether" step; Kunin et al.'s Theorem 1 states the general condition). Any function of c is also conserved. The argument used only that L is invariant under (2), which holds for every $L(a,b) = \ell(ab)$: then $\nabla L = \ell'(ab)(b,a)$ and $\langle \nabla L, X\rangle = \ell'(ab)(ab - ba) = 0$. This covers any per-example loss of $f_{a,b}(x) = abx$ on any dataset (squared, logistic, cross-entropy), since the data enter only through the scalar ab.

(b) Differentiate Equation (S1), $\langle \nabla L(\theta), X(\theta)\rangle = 0$, with respect to θ. The product rule gives

$$\nabla^2 L(\theta)\, X(\theta) + (\partial X/\partial\theta)^\top \nabla L(\theta) = 0, \quad \text{here} \quad \nabla^2 L\, X = -\mathrm{diag}(1,-1)\nabla L. \tag{S3}$$

At any minimum $\nabla L = 0$, so $\nabla^2 L\, X = 0$: the generator is a zero-eigenvalue direction of the Hessian. On $ab = 1$, $X = (a, -1/a) = a(1, -1/a^2)$, which is exactly A0′s zero-eigenvalue eigenvector, the tangent to the curve of minima. The symmetry *is* the flat direction: moving along the orbit of ψ_α moves along the minima without changing the loss.

(c) With a_0, b_0 i.i.d. $\mathcal{N}(0,\sigma^2)$, $\mathbb{E}[a^4] = 3\sigma^4$, so

$$\mathbb{E}[c^2] = \mathbb{E}[a^4] - 2\mathbb{E}[a^2]\mathbb{E}[b^2] + \mathbb{E}[b^4] = 6\sigma^4 - 2\sigma^4 = 4\sigma^4, \quad \sqrt{\mathbb{E}[c^2]} = 2\sigma^2. \tag{S4}$$

At the endpoint $a_\infty^2 - b_\infty^2 = (a_\infty - b_\infty)(a_\infty + b_\infty) = c$ and $(a_\infty + b_\infty)^2 = a_\infty^2 + b_\infty^2 + 2 = \sqrt{c^2+4} + 2 = 4 + O(c^2)$, so

$$a_\infty - b_\infty = \frac{c}{\sqrt{\sqrt{c^2+4}+2}} = \frac{c}{2} + O(c^3). \tag{S5}$$

The init scale enters squared: an init of scale 10^{-2} leaves the two layers equal to about 10^{-4} at the end of training. This is Du, Hu and Lee's "layers are automatically balanced".

(d) Reference: `solutions/noether_solution.py` (three one-liners on `theta[..., 0]`, `theta[..., 1]`).

(e) Expected. $|\langle \nabla L, X\rangle| \,/(\|\nabla L\|\|X\|) \approx 1.3 \times 10^{-16}$ for the three losses of ab and 0.998 for the skip connection. $\max_t |Q(t) - Q(0)|$ on $[0, 10]$: 5.0×10^{-11}, 1.4×10^{-10}, 3.2×10^{-11} (integrator error) for the three flows, 0.103 for the skip connection and 0.096 for GD at $\eta = 0.1$. Hessian identity residual 1.0×10^{-9} (finite-difference error), $\|\nabla^2 L\, X\| = 4.6 \times 10^{-16}$ on $ab = 1$. Root-mean-square c: $2.03, 18.1, 202, 1811 (\times 10^{-4})$ and 1.98 against $2\sigma^2$; the endpoint gap matches $c/2$ to 0.5% for $\sigma \le 0.3$ and is 0.66 against 0.80 at $\sigma = 1$, where c is not small.

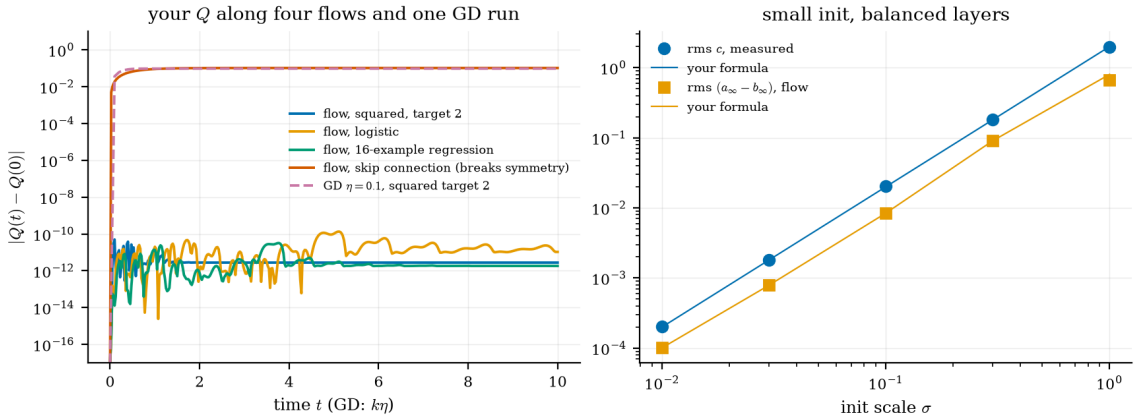


Figure 1: `noether.png`. Left: $|Q(t) - Q(0)|$ along four flows and one GD run. Right: the size of c and of the endpoint gap against the init scale.

2 Problem (`endpoint`): Where gradient flow lands

(a) Write $A = a_\infty^2$. Conservation gives $b_\infty^2 = A - c$ and the endpoint gives $a_\infty^2 b_\infty^2 = s^2$, so $A^2 - cA - s^2 = 0$. Only the root with $+$ is positive:

$$a_\infty^2 = \frac{c + \sqrt{c^2 + 4s^2}}{2}, \quad b_\infty^2 = \frac{-c + \sqrt{c^2 + 4s^2}}{2}. \tag{S6}$$

Signs: if $c > 0$ then $a^2 = c + b^2 \ge c > 0$ for all t, so a never crosses zero and keeps the sign of a_0; b_∞ has the same sign since $a_\infty b_\infty = s > 0$. If $c < 0$, likewise b keeps its sign. In both cases $\operatorname{sign}(a_0) = \operatorname{sign}(a_0 + b_0)$ (resp. $\operatorname{sign}(b_0)$), because the larger of $|a_0|, |b_0|$ wins the sum. If $c = 0$ the start is on $a = b$ (stays there, ends at $\pm\sqrt{s}$ with the sign of $a_0 + b_0$) or on $a = -b$ (A0: flows to the origin). One rule covers everything:

$$(a_\infty, b_\infty) = \operatorname{sign}(a_0 + b_0) \left(\sqrt{\left(c + \sqrt{c^2 + 4s^2}\right)/2}, \sqrt{\left(-c + \sqrt{c^2 + 4s^2}\right)/2} \right), \tag{S7}$$

with $\operatorname{sign}(0) = 0$ giving the origin. (Convergence to $ab = s$ for $c \neq 0$: $\dot{p} = \sqrt{c^2 + 4p^2}(s - p)$ with a strictly positive square root; see Problem (`logistic`) (e).)

(b) Reference: `solutions/endpoint_solution.py`.

(c) Expected. Maximum error 1.0–1.2×10^{-10} over 150 starts for each of $s = 0.5, 1, 2$. From $(0.7, -0.7 + \varepsilon)$, the loss falls below $1/4$ at $t = 4.67, 9.27, 13.87$ for $\varepsilon = 10^{-2}, 10^{-4}, 10^{-6}$: each factor 100 in ε adds $4.6 = \ln 100$. The start sits near the origin's stable direction $a = -b$ (eigenvalue -1), is pulled to the saddle, and leaves along $a = b$ at rate $+1$ starting from a component of size about ε, so it needs time $\ln(1/\varepsilon)$.

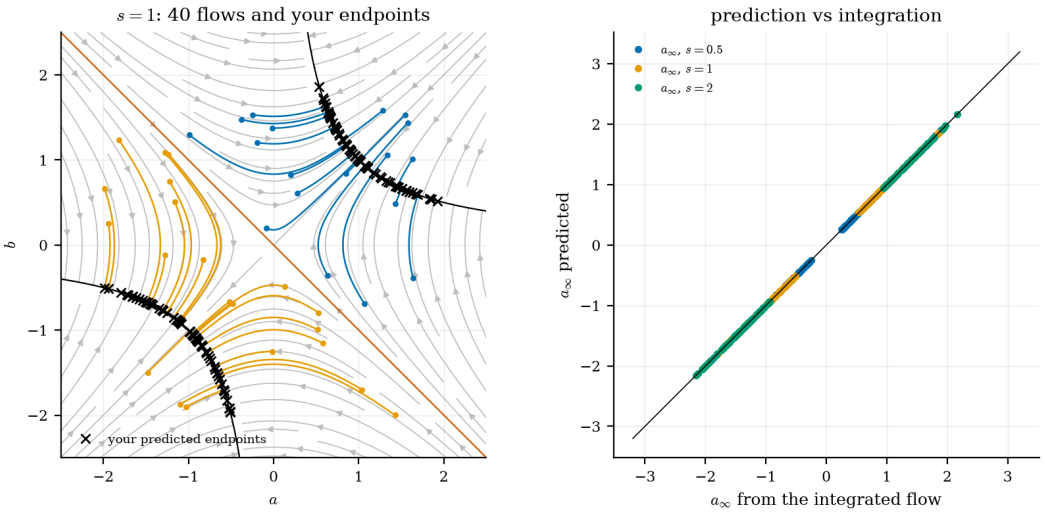


Figure 2: `endpoint.png`. Left: 40 flows for $s = 1$, coloured by $\text{sign}(a_0 + b_0)$, with the predicted endpoints. Right: predicted against integrated a_∞ for all 450 starts.

3 Problem (`logistic`): Balanced learning is logistic

(a) For $L = \frac{1}{2}(s - ab)^2$: $\dot{a} = b(s - ab)$, $\dot{b} = a(s - ab)$, so $\frac{d}{dt}(a - b) = -(a - b)(s - ab)$. This is linear in $a - b$; with $a_0 - b_0 = 0$ the unique solution is $a - b \equiv 0$. On $a = b$, $u = ab = a^2$ and

$$\dot{u} = \dot{a}b + a\dot{b} = (a^2 + b^2)(s - u) = 2u(s - u). \tag{S8}$$

Separate and use partial fractions, $\int du/(u(s-u)) = \frac{1}{s}\ln(u/(s-u))$:

$$\frac{1}{2s}\ln\frac{u(s-u_0)}{u_0(s-u)} = t \quad \Rightarrow \quad u(t) = \frac{su_0e^{2st}}{s + u_0(e^{2st} - 1)} = \frac{s}{1 + (s/u_0 - 1)e^{-2st}}. \tag{S9}$$

The second form never overflows. This is Saxe et al.'s eq. (12) with $\tau = 1$.

(b) Solving Equation (S9) for $u(t_f) = fs$:

$$t_f(u_0, s) = \frac{1}{2s}\ln\frac{f(s-u_0)}{(1-f)u_0}, \quad t_{1/2} = \frac{1}{2s}\ln\frac{s-u_0}{u_0} = \frac{1}{2s}\ln\frac{s}{u_0} + O(u_0/s). \tag{S10}$$

Plateau (time to reach $0.1s$): $\frac{1}{2s}\ln(s/(9u_0))$, logarithmic in $1/u_0$ and inversely proportional to s. *Drop* (from $0.1s$ to $0.9s$): $\frac{1}{2s}(\ln 9 - \ln(1/9)) = \ln 81/(2s) = 2.20/s$, independent of u_0. For $s = 1$, $u_0 = 10^{-6}$: plateau 5.82, drop 2.20. Smaller init lengthens only the plateau, so the curve looks ever more like a step.

(c) Reference: `solutions/logistic_solution.py`.

(d) Expected. Maximum relative error of $t_{1/2}$ over 60 pairs: 1.2×10^{-10}. Fitted slopes of $t_{1/2}$ against $\ln(1/u_0)$: $1.0065, 0.5015, 0.2504, 0.1251$ for $s = 0.5, 1, 2, 4$, i.e. $1/(2s)$ (the small excess at $s = 0.5$ is the $O(u_0/s)$ term at $u_0 = 0.1$).

(e) Expected prediction: faster. In general $\dot{p} = \dot{a}b + a\dot{b} = (a^2 + b^2)(s - p)$ and $(a^2 + b^2)^2 = (a^2 - b^2)^2 + 4a^2b^2$, so

$$\dot{p} = \sqrt{c^2 + 4p^2}\,(s - p). \tag{S11}$$

For $p \ll |c|$ the rate is about $|c|$ s, a constant push, instead of $2ps$, which is tiny at small p; the plateau is the time the balanced net spends with a vanishing rate. Imbalance matters once $|c|$ is comparable to $2p_0 = 2 \times 10^{-4}$. Measured half-times at $p_0 = 10^{-4}$, $s = 1$: 4.61 ($c = 0$, $= \ln(10^4)/2$),

$4.58, 4.05, 2.98, 2.44, 1.82, 1.23, 0.60$ for $c = 10^{-4}, 10^{-3}, 10^{-2}, 0.03, 0.1, 0.3, 1$. The kernel of the anchor is $a^2 + b^2 \geq |c|$ (A5), so imbalance also keeps it from starting at zero.

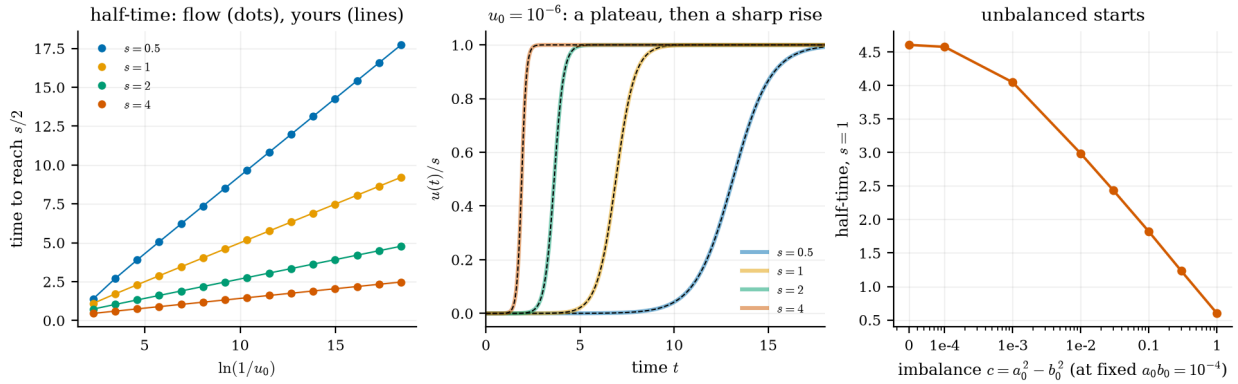


Figure 3: `logistic.png`. Left: half-time against $\ln(1/u_0)$, slopes $1/(2s)$. Middle: $u(t)/s$ for $u_0 = 10^{-6}$. Right: half-time against the imbalance at fixed $a_0 b_0 = 10^{-4}$.

4 Problem (`matrix-law`): The matrix conservation law

(a) $L(W) = \frac{1}{2}\operatorname{tr}(W\Sigma_{xx}W^\top) - \operatorname{tr}(W\Sigma_{yx}^\top)$ has $G = \nabla_W L = W\Sigma_{xx} - \Sigma_{yx}$. Perturbing $W_1 \to W_1 + E$ changes W by $W_2 E$ and L by $\operatorname{tr}(G^\top W_2 E) = \langle W_2^\top G, E\rangle$ to first order; perturbing $W_2 \to W_2 + E$ changes W by EW_1 and L by $\langle GW_1^\top, E\rangle$:

$$\nabla_{W_1} L = W_2^\top G, \quad \nabla_{W_2} L = GW_1^\top. \tag{S12}$$

(b) Under gradient flow $\dot{W}_1 = -W_2^\top G$, $\dot{W}_2 = -GW_1^\top$, so

$$\begin{aligned} \dot{D} &= \dot{W}_1 W_1^\top + W_1 \dot{W}_1{}^\top - \dot{W}_2{}^\top W_2 - W_2^\top \dot{W}_2 \\ &= -W_2^\top G W_1^\top - W_1 G^\top W_2 + W_1 G^\top W_2 + W_2^\top G W_1^\top = 0. \end{aligned} \tag{S13}$$

Only the form of the two gradients was used, so any loss of the product $W_2 W_1$ works. Noether-style: $L(AW_1, W_2 A^{-1})$ is constant; with $A = I + \varepsilon M$, $A^{-1} = I - \varepsilon M + O(\varepsilon^2)$, and differentiating at $\varepsilon = 0$,

$$\left\langle \nabla_{W_1} L, MW_1 \right\rangle - \left\langle \nabla_{W_2} L, W_2 M \right\rangle = \operatorname{tr}\Big(M\big(W_1 \nabla_{W_1} L^\top - \nabla_{W_2} L^\top W_2\big)\Big) = 0 \quad \text{for all } M, \tag{S14}$$

which forces the $m \times m$ matrix $W_1 \nabla_{W_1} L^\top - \nabla_{W_2} L^\top W_2$ to vanish (here both terms equal $W_1 G^\top W_2$). Then $\dot{D} = -\big(\nabla_{W_1} L\, W_1^\top + W_1 \nabla_{W_1} L^\top\big) + \big(\nabla_{W_2} L^\top W_2 + W_2^\top \nabla_{W_2} L\big) = 0$ by the constraint and its transpose: one conserved number per entry of D, one per generator M. For $m = d_{\text{in}} = d_{\text{out}} = 1$, M is a scalar, the constraint is $a\partial_a L - b\partial_b L = 0$, and $D = a^2 - b^2$.

(c) With $g_i = \nabla_{W_i} L$: $W_{1'} W_{1'}^\top = W_1 W_1^\top - \eta(g_1 W_1^\top + W_1 g_1^\top) + \eta^2 g_1 g_1^\top$ and $W_{2'}^\top W_{2'} = W_2^\top W_2 - \eta(g_2^\top W_2 + W_2^\top g_2) + \eta^2 g_2^\top g_2$. The two first-order terms are both $W_2^\top G W_1^\top + W_1 G^\top W_2$, so they cancel *exactly*, and

$$D' - D = \eta^2\big(\nabla_{W_1} L\, \nabla_{W_1} L^\top - \nabla_{W_2} L^\top \nabla_{W_2} L\big). \tag{S15}$$

Scalar case: $\eta^2(b^2 r^2 - a^2 r^2) = -\eta^2 r^2 c$, i.e. $c' = c(1 - \eta^2 r^2)$, the post's §3.1 formula.

(d) Reference: `solutions/two_layer_solution.py`.

(e) Expected. Flow drift 5.3×10^{-13}; Equation (S15) matches every GD step to 1.1×10^{-15}; drift at $t = 10$: $0.261, 0.115, 0.0545, 0.0212, 0.0105, 0.00521$ for $\eta = 0.2, \ldots, 0.005$, fitted slope 1.05. Prediction $p = 1$: each step adds $O(\eta^2)$ and reaching $t = 10$ takes $10/\eta$ steps.

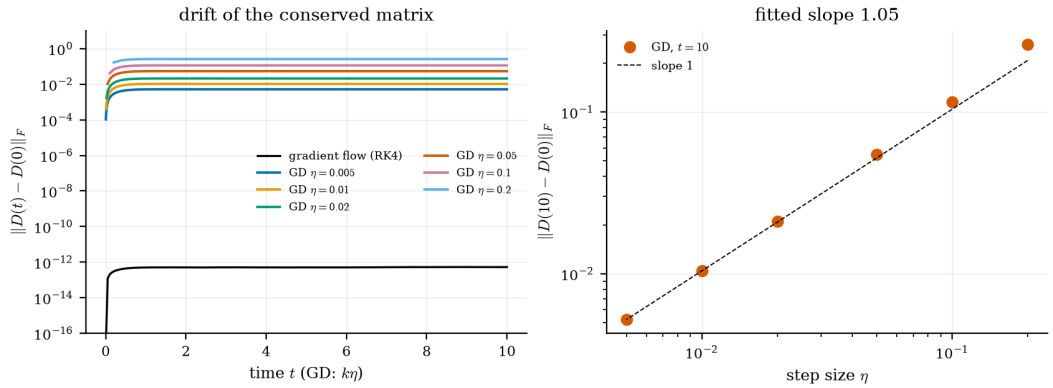


Figure 4: `matrix_law.png`: drift of D along the flow and GD (left), and at $t = 10$ against η (right).

5 Problem (saxe): Deep linear networks learn mode by mode

(a) With $\Sigma_{xx} = I$ the flow is $\dot{W}_1 = W_2^\top(\Sigma_{yx} - W_2W_1)$, $\dot{W}_2 = (\Sigma_{yx} - W_2W_1)W_1^\top$. Multiply by the orthogonal $U^\top$, V: with $S = U^\top\Sigma_{yx}V$ (diagonal, s_α on the diagonal),

$$\dot{\overline{W}}_1 = \overline{W}_2^\top\big(S - \overline{W}_2\overline{W}_1\big), \quad \dot{\overline{W}}_2 = \big(S - \overline{W}_2\overline{W}_1\big)\overline{W}_1^\top. \tag{S16}$$

Ansatz: $\overline{W}_1 = R\,\mathrm{diag}(a_\alpha)$, $\overline{W}_2 = \mathrm{diag}(b_\alpha)R^\top$ (padded with zeros for the other directions). Then $\overline{W}_2\overline{W}_1 = \mathrm{diag}(b_\alpha)R^\top R\,\mathrm{diag}(a_\alpha) = \mathrm{diag}(a_\alpha b_\alpha)$ because $R^\top R = I$, so $S - \overline{W}_2\overline{W}_1 = \mathrm{diag}(s_\alpha - a_\alpha b_\alpha)$ and

$$\dot{\overline{W}}_1 = R\,\mathrm{diag}(b_\alpha(s_\alpha - a_\alpha b_\alpha)), \quad \dot{\overline{W}}_2 = \mathrm{diag}(a_\alpha(s_\alpha - a_\alpha b_\alpha))R^\top. \tag{S17}$$

The velocity has the ansatz's form, so (by uniqueness of solutions) the ansatz holds for all time with

$$\dot{a}_\alpha = b_\alpha(s_\alpha - a_\alpha b_\alpha), \quad \dot{b}_\alpha = a_\alpha(s_\alpha - a_\alpha b_\alpha), \tag{S18}$$

the anchor with target s_α. Modes do not interact because the cross terms $r_\alpha^\top r_\beta$ vanish (Saxe's competitive terms in their eq. (5) are zero on this manifold). Undoing the rotation, $W = U\,\mathrm{diag}(a_\alpha b_\alpha)V^\top$, which is (6).

(b) The init is balanced ($a_\alpha = b_\alpha = \sqrt{u_{0\alpha}}$), so each mode is Problem (`logistic`):

$$u_\alpha(t) = \frac{s_\alpha}{1 + (s_\alpha/u_{0\alpha} - 1)e^{-2s_\alpha t}}, \quad \frac{t_{1/2,\alpha}}{t_{1/2,\beta}} = \frac{s_\beta \ln(s_\alpha/u_0)}{s_\alpha \ln(s_\beta/u_0)} \approx \frac{s_\beta}{s_\alpha}. \tag{S19}$$

Strong modes are learned first, at a time inversely proportional to their singular value (up to the logarithm). In the 2×2 example the "subtract" mode comes about 3 times later.

(c) Reference: `solutions/saxe_solution.py` (`R` = the first r columns of I_m).

(d) Linearize: $\dot{x} = \dot{W}_1\boldsymbol{v}_\alpha = W_2^\top(\Sigma_{yx} - W)\boldsymbol{v}_\alpha \approx W_2^\top\Sigma_{yx}\boldsymbol{v}_\alpha = s_\alpha W_2^\top\boldsymbol{u}_\alpha = s_\alpha y$, and likewise $\dot{y} = \dot{W}_2{}^\top\boldsymbol{u}_\alpha \approx W_1\Sigma_{yx}^\top\boldsymbol{u}_\alpha = s_\alpha x$. Hence $x + y = (x_0 + y_0)e^{s_\alpha t}$ and $x - y = (x_0 - y_0)e^{-s_\alpha t}$, and

$$u_\alpha = x^\top y = \frac{1}{4}(\|x + y\|^2 - \|x - y\|^2) \approx \frac{1}{4}\|x_0 + y_0\|^2 e^{2s_\alpha t}, \quad u_0^{\mathrm{eff}} = \frac{1}{4}\big\|W_1(0)\boldsymbol{v}_\alpha + W_2(0)^\top\boldsymbol{u}_\alpha\big\|^2. \tag{S20}$$

For entries of variance σ^2 this is about $m\sigma^2/2$. A random init does not start on the aligned manifold, but the growing direction $x + y$ aligns it within the plateau; the decaying part is forgotten.

(e) Expected. Aligned init: $\max|W_{\mathrm{flow}} - W_{\mathrm{pred}}| = 1.7 \times 10^{-9}$ and half-times $1.900, 3.627, 6.908, 13.122$ equal to the formula. Random init ($u_0^{\mathrm{eff}} = 5.3, 5.3, 1.4, 2.0 \times 10^{-6}$): measured half-times $1.691, 3.223, 6.852, 12.792$ against predicted $1.691, 3.209, 6.725, 12.423$, relative delays

$0.0\%, 0.4\%, 1.9\%, 3.0\%$, growing for the weaker modes as in Saxe et al.'s Fig. 3 (right). $s_\alpha t_{1/2}$ stays between 6.4 and 6.9: the time scale is $1/s_\alpha$, the rest is the logarithm.

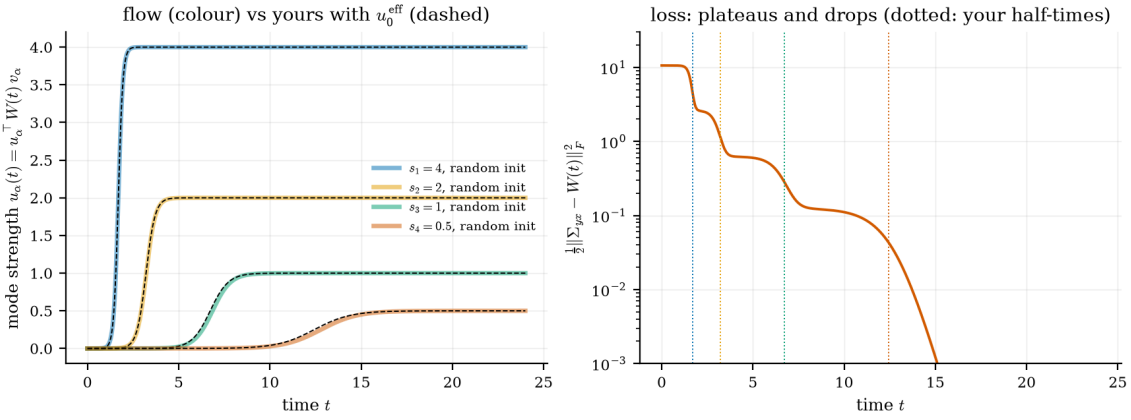


Figure 5: `saxe.png`. Left: mode strengths from a random init against the prediction with u_0^{eff}. Right: the loss, a staircase of plateaus and drops.

6 Problem (`broken`): Finite steps, momentum, minibatches and weight decay

(a) Write (7) as $a' = (1-2\eta\gamma)a - \eta b r$, $b' = (1-2\eta\gamma)b - \eta a r$. Then

$$a'^2 - b'^2 = (1-2\eta\gamma)^2(a^2-b^2) - 2(1-2\eta\gamma)\eta r(ab-ba) + \eta^2 r^2(b^2-a^2), \tag{S21}$$

$$c_{k+1} = c_k\big[(1-2\eta\gamma)^2 - \eta^2 r_k^2\big]. \tag{S22}$$

With $\gamma = 0$, $|c|$ cannot grow iff $|1-\eta^2 r_k^2| \le 1$, i.e. $\eta\ |r_k| \le \sqrt{2}$; for $\eta\ |r_k| > \sqrt{2}$ the factor is below -1 and c flips sign and grows (the post's diverging example). Under the flow on L_γ, $\dot a = -br - 2\gamma a$, $\dot b = -ar - 2\gamma b$, and

$$\dot c = 2a\dot a - 2b\dot b = -4\gamma c, \quad c(t) = c_0 e^{-4\gamma t}. \tag{S23}$$

(The brief's convention $\dot a = -\partial_a L - \gamma a$ gives $-2\gamma c$; Kunin et al.'s $L + \lambda/2\|\theta\|^2$ gives $-2\lambda c$, App. E.) Expanding Equation (S22): $(1-2\eta\gamma)^2 \approx e^{-4\eta\gamma}$, the flow's decay over a time η.

(b) Along (8), $\dot c = -2\langle X, \nabla L\rangle - \eta\langle X, \nabla^2 L \nabla L\rangle$ with $X = (a, -b)$. The first term vanishes by Equation (S1). For the second, Equation (S3) gives $\langle X, \nabla^2 L \nabla L\rangle = \langle \nabla^2 L X, \nabla L\rangle = -\langle \mathrm{diag}(1,-1)\nabla L, \nabla L\rangle = -(b^2 r^2 - a^2 r^2) = r^2 c$. So

$$\dot c = -\eta r^2 c. \tag{S24}$$

(Directly: $\nabla^2 L \nabla L = r(b^3 + (2ab-1)a, (2ab-1)b + a^3)$, dotted with $(a,-b)$ gives $r(ab(a^2-b^2) - (a^2 - b^2)) = r^2 c$.) This is Kunin et al.'s rescale result (App. E, eq. (30)) with $|g_a|^2 - |g_b|^2 = r^2(b^2 - a^2)$.

(c) Dividing Equation (S24) by c and integrating, $\ln(c_\infty/c_0) = -\eta \int_0^\infty r^2\, dt$; to first order in η the integral may be taken along gradient flow. There $\dot p = \sqrt{c^2 + 4p^2}(1-p) = -r\sqrt{c^2+4p^2}$, monotone from p_0 to 1, so $r^2 dt = r^2 dp/\dot p = (1-p)dp/\sqrt{c^2+4p^2}$ and

$$I(p_0, c) = \int_{p_0}^1 \frac{1-p}{\sqrt{c^2+4p^2}} dp = \frac{1}{2}\left[\operatorname{arcsinh}\left(\frac{2}{|c|}\right) - \operatorname{arcsinh}\left(\frac{2p_0}{|c|}\right)\right] - \frac{1}{4}\left[\sqrt{c^2+4} - \sqrt{c^2+4p_0^2}\right], \tag{S25}$$

$$\ln(c_\infty/c_0) \approx -\eta\, I(p_0, c_0). \tag{S26}$$

Scaling: one step changes $\ln c$ by $\ln(1-\eta^2 r_0^2) \approx -\eta^2 r_0^2$, slope 2 in η; a whole run takes about $1/\eta$ steps (to a fixed time), so the total is $O(\eta)$, slope 1. The drift is always negative: finite steps shrink $|c|$, pushing toward balance, and more so the longer the run spends with large r (for $p_0, |c| \ll 1$, $I \approx \frac{1}{2}\ln(4/|c|) - \frac{1}{2}$, which grows as $c \to 0$).

(d) Take the inner product of (9) with $X = (a, -b)$. Since $\left\langle X, \dot{\theta} \right\rangle = \dot{c}/2$, $\left\langle X, \ddot{\theta} \right\rangle = \frac{1}{2}\ddot{c} - \left(\dot{a}^2 - \dot{b}^2\right)$ and $\langle X, \nabla L \rangle = 0$:

$$\eta \frac{1+\beta}{2}\left[\frac{1}{2}\ddot{c} - \left(\dot{a}^2 - \dot{b}^2\right)\right] + \frac{1-\beta}{2}\dot{c} = 0. \tag{S27}$$

To leading order $\dot{\theta} = -\nabla L/(1-\beta)$, so $\dot{a}^2 - \dot{b}^2 = r^2(b^2 - a^2)/(1-\beta)^2 = -r^2 c/(1-\beta)^2$, and

$$\dot{c} = -\eta \frac{1+\beta}{(1-\beta)^3} r^2 c - \eta \frac{1+\beta}{2(1-\beta)} \ddot{c} + O(\eta^2). \tag{S28}$$

The last term integrates to $\dot{c}(\infty) - \dot{c}(0) = 0$ ($\dot{\theta}(0) = 0$). In t, the leading-order trajectory is gradient flow run at speed $1/(1-\beta)$, so $\int r^2 dt = (1-\beta) I(p_0, c_0)$. Hence

$$\ln(c_\infty / c_0) \approx -\eta \frac{1+\beta}{(1-\beta)^2} I(p_0, c_0), \tag{S29}$$

a factor $(1+\beta)/(1-\beta)^2$ over GD at the same η; in terms of the effective step $\eta_{\text{eff}} = \eta/(1-\beta)$ it is $(1+\beta)/(1-\beta)$ times GD at η_{eff}. Momentum breaks the law more than plain GD with the same effective step, because of the inertia term.

(e) Per-example gradient: $\partial_a \frac{1}{2}(abx_i - x_i)^2 = bx_i^2 r$, so the batch gradient is $m_B r(b, a)$ with $m_B = \frac{1}{B}\sum_{i \in B} x_i^2$: a multiple of (b, a), orthogonal to X. One step is the GD step with η replaced by ηm_B:

$$c_{k+1} = c_k(1 - \eta^2 m_B^2 r_k^2). \tag{S30}$$

To first order the drift is $-\eta^2 \sum_k \mathbb{E}[m_B^2] r_k^2 \approx -\eta \mathbb{E}[m_B^2] I$ (the mean path is the flow since $\mathbb{E}[m_B] = 1$), and for B draws with replacement $\mathbb{E}[m_B^2] = 1 + \mathrm{Var}(x_i^2)/B$. So SGD multiplies the drift by $1 + \mathrm{Var}(x^2)/B$; for Gaussian x that is $1 + 2/B$.

(f) Reference: `solutions/drift_solution.py`.

(g) Expected. (A) Equation (S22) exact to 9×10^{-16}. (B) weight decay: relative error 2×10^{-11} ($\gamma = 0.01$) and 2×10^{-8} ($\gamma = 0.05$, after c has decayed by e^{-8}). (C) $I(0.03, 0.08) = 1.13424$; slopes 2.001 (one step) and 1.015 (whole run). Measured over predicted $\ln(c_\infty/c_0)$:

η	0.1	0.05	0.02	0.01	0.005	0.002
GD	1.064	1.031	1.012	1.006	1.003	1.001
$\beta = 0.3$	1.164	1.070	1.026	1.012	1.006	1.002
$\beta = 0.5$	1.603	1.186	1.059	1.027	1.013	1.005

Table 1: The ratio tends to 1 linearly in η (the neglected $O(\eta^2)$ term), faster for GD than for momentum, whose effective step is larger.

(E) dataset with $\mathrm{Var}(x_i^2) = 3.24$: SGD ratio 4.28 ± 0.06, 1.83 ± 0.01, 1.198 ± 0.005 for $B = 1, 4, 16$ against $4.24, 1.81, 1.20$.

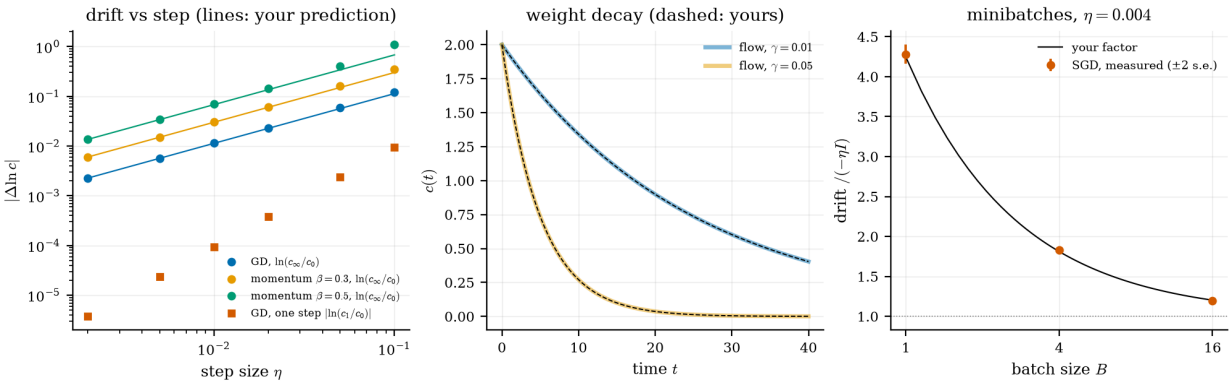


Figure 6: `drift.png`. Left: $|\Delta \ln c|$ against η for GD and momentum (one run each to convergence) and for a single GD step. Middle: $c(t)$ under weight decay. Right: SGD's drift relative to $-\eta I$ against batch size.

7 Problem (`framing`): reading notes

No single right answer; a good answer covers the following. (a) The second run of `check_saxe.py` dropped the task-aligned initialization (inputs stay whitened). The cost: one needs u_0^{eff} instead of the init scale, and the half-times come 0 to 3% late, more for weaker modes, from competition between modes during the plateau. Simon et al.'s Fig. 1(a) uses the aligned init, where the prediction is exact. (b) Predicted: GD's exact one-step factor and its $O(\eta)$ total drift, momentum's factor $(1+\beta)/(1-\beta)^2$, SGD's factor $1 + \mathrm{Var}/B$, weight decay's exact exponential. Not covered: steps so large that the run is far from the flow (A3′s edge of stability, where ηr is not small), label noise and other noise that does not vanish at the minimum (A4), adaptive optimizers such as Adam, whose per-coordinate rescaling makes the update no longer a multiple of a gradient orthogonal to X, and normalization layers that add other symmetries. (c) Scores depend on $xW_QW_K^\top x'^\top$, invariant under $W_Q \to W_QA$, $W_K \to W_KA^{-\top}$ for invertible A. With $W_2 = W_Q$, $W_1 = W_K^\top$ this is Problem (`matrix-law`), and gradient flow conserves $W_K^\top W_K - W_Q^\top W_Q$ (Simon et al. §2.3; Kunin et al. treat such symmetries in §3).